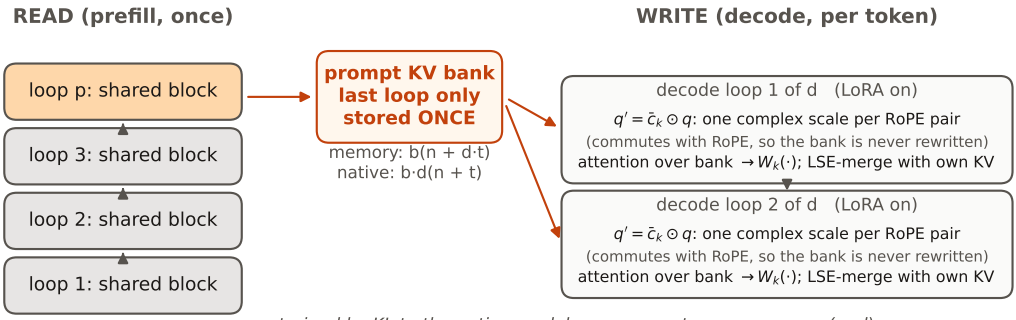
READ (prefill, once)
WRITE (decode, per token)
loop p: shared block
loop 3: shared block
loop 2: shared block
loop 1: shared block
prompt (n tokens), p loops
prompt KV bank
last loop only
stored ONCE
memory: b(n + d·t)
native: b·d(n + t)
decode loop 1 of d (LoRA on)
$q' = \bar{c}_k \odot q$: one complex scale per RoPE pair
(commutes with RoPE, so the bank is never rewritten)
attention over bank → $W_k(\cdot)$; LSE-merge with own KV
decode loop 2 of d (LoRA on)
$q' = \bar{c}_k \odot q$: one complex scale per RoPE pair
(commutes with RoPE, so the bank is never rewritten)
attention over bank → $W_k(\cdot)$; LSE-merge with own KV
trained by KL to the native model; one connector serves every (p, d)